

# Tunisia, audited — Cut the Dose

The burden measured, the cheapest cut in each chain, the 90-day pilot, the public scoreboard, and the food stack — plain words, full arithmetic.

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## 1. The burden, measured — and what the numbers say

Roughly **379 years of healthy life lost every year** in the basin (30-70k people): 200 to fluoride — teeth and bones — 120 to heavy metals, 59 to dust (H4-H6 anchors, DERIVED). A child in Redeyef develops asthma at age two; the nearest hospital is 70 km; the doctors' prescription is *leave Gafsa* (H16).

## 2. Cut the Dose — the cheapest link in each chain

| CHAIN                           | THE CUT                                                              | COST                                                                                |
|---------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Fluoride: water → teeth → bones | point-of-use filters (Nalgonda/bone char)                            | <b>\$2,592 per healthy year</b> vs the \$4,000 standard — this rung pays for itself |
| Metals: water → gut → kidney    | zinc, selenium, probiotics (gut binding)                             | in the \$2.20M/yr full stack                                                        |
| Dust: air → lungs               | clean-air boxes + N95-class masks (98%+ vs ≤44% cloth)               | RCT: poorly-controlled asthma IRR 0.43-0.45                                         |
| Fluorosis in children           | calcium + vitamin C + D3 — the combination shown to REVERSE it (RCT) | pennies a day                                                                       |

## 3. The 90-day pilot — and the scoreboard that follows

**\$58,000:** 100 households + 2 schools, 90 days, results published. **Gate:** filter uptime ≥70%, urinary F ↓20%. **The scoreboard:** water chemistry, filter uptime, cohort KPIs — public, every quarter. The health data of the basin becomes the scoreboard of the nation — and the collateral that lowers the price of every loan in the plan (100-150bp on ~\$800M of portfolio debt: \$8-12M/yr).

## 4. The food stack — protein the people can pay

| PIECE           | NUMBER                                                                                       |
|-----------------|----------------------------------------------------------------------------------------------|
| Spirulina ponds | 10 ha → 130 t/yr → <b>178,082 children</b> at the RCT dose (2 g/day); the basin needs 0.7 ha |
| Protein price   | \$5-10/kg where imported meat protein costs \$15-25/kg                                       |

| PIECE                 | NUMBER                                                                                                                        |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Terfas domestic quota | $\geq 30\%$ of harvest sold at home at $\leq \text{€}12/\text{kg}$ — written into the contracts                               |
| Bone char             | 90 t/yr from local slaughterhouses → the filters, and a workshop of jobs                                                      |
| Success metric        | the price of protein and vegetables in Gafsa and Gabès bends downward within 24 months of first harvest — published quarterly |

## 5. The two futures, in plain words

**Without the plan:** the dose accumulates — fluoride in the teeth before age 8, cadmium in the kidneys for decades, dust in the lungs. The fittest leave: engineer to Paris, founder to London. Every departure makes the basin harder for those who stay. That future requires no decision — it is the default.

**With the plan:** we change the environment, not the genes. Clean water → intact enamel and strong bones within one generation. Protein → nourished brains. Jobs → a reason to stay. Same people, same geography — a different fitness landscape. The plan is evolution by other means.

**What this document is:** a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

**What it is not:** a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

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